

ECON8037: Financial Economics

Week 2 Lecture - Introduction to Financial Economics

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Information Efficiency

In this lecture material we will look at the effect that information has on asset prices.

When we discuss informational efficiency, we are asking how much information related to an asset's value is incorporated in the price of that asset. This idea is summarised in [Fama \(1970\)](#).

The degree of efficiency in some sense determines which assets to hold. For example, if asset markets are perfectly informationally efficient then an investor can do no better than to hold the market portfolio, which has the highest return relative to its volatility.

Information Efficiency

There are three notions of informational efficiency, these are different variations of what is known as the [Efficient Markets Hypothesis \(EMH\)](#)

- ① [Strong form](#) - all information in a market, whether public or private, is accounted for in an asset's price
- ② [Semi-strong form](#) - all public (but not non-public) information is calculated in an asset's current share price
- ③ [Weak form](#) - all past prices of an asset are reflected in the current asset price

Three Puzzles

If asset markets are perfectly efficient in the sense of Fama (1970), then three puzzles emerge (the ones in blue will be investigated this lecture, the one in red will be looked at in future lectures)

- ① First, no one has an incentive to collect information in an efficient market, so how does the market become efficient? – This is commonly referred to as the Grossman and Stiglitz (1980) paradox
- ② Second, if asset markets are efficient, then positive fees to active managers implies an inefficient market for asset management – Gârleanu and Pedersen (2018)
- ③ Third, why do asset prices still move in the absence of new publicly available information?

Empirical Regularities

There exists a large literature on the performance of asset managers

- The ‘old consensus’ was that the average asset manager has no skill ([Fama \(1970\)](#) and [Carhart \(1997\)](#)) – no superior risk adjusted returns
- A ‘new consensus’ has emerged that the average hides significant cross-sectional variation in manager skill ([Grinblatt and Titman \(1989\)](#) and [Kosowski et al. \(2007\)](#))
 - “a sizeable minority of managers pick stocks well enough to more than cover their costs”
- Another empirical observation is that search costs have diminished over time as information technology has improved and as a result markets have become more efficient as per [Wurgler \(2000\)](#) and [Bai et al. \(2016\)](#)
 - And this is linked to the amount of assets managed by professional investors ([Rösch et al. \(2015\)](#))

We will look at a model by [Gârleanu and Pedersen \(2018\)](#) now that can account for the first two puzzles and these empirical regularities.

Gârleanu and Pedersen

Consider a setting where searching investors trade directly or through asset managers, asset managers trade on behalf of groups of investors, noise allocators make random allocations to asset managers, and noise traders make random trades in financial markets.

The economy has $\bar{A}$ searching investors (“allocators”), each of whom can

- ① Invest directly in asset markets after having acquired a signal s at cost k
- ② Invest directly in asset markets without the signal, or
- ③ Invest through an asset manager

Gârleanu and Pedersen

Due to economies of scale, a natural equilibrium outcome is that investors do not acquire the signal, but rather invest as uninformed or through a manager. We therefore rule out that investors acquire the signal. Consequently, we focus on

- ① The number A of investors who make informed investments through a manager
- ② Therefore the number of uninformed investors is $\bar{A} - A$

The economy has $\bar{M}$ risk-neutral asset management firms. Of these

- ① Only M elect to pay k to acquire the signal s and therefore become informed asset managers
- ② The remaining $\bar{M} - M$ managers seek to collect asset management fees and invest without information

Gârleanu and Pedersen

To invest with an informed asset manager, investors must search for and vet managers, which is costly. Specifically, the cost of finding an informed manager and confirming that they have the signal is given by the general continuous function $c(M, A)$, which obviously depends on the number of informed asset managers M and the number of their investors A .

Gârleanu and Pedersen

We assume that all investors have constant absolute risk aversion (CARA) utility over end-of-period consumption with risk-aversion parameter γ . For convenience, we express the utility as certainty-equivalent wealth — hence, with end-date wealth $\tilde{W}$, an investor's utility is

$$-\frac{1}{\gamma} \ln \left(\mathbb{E} \left[e^{-\gamma \tilde{W}} \right] \right)$$

Each investor is endowed with initial wealth W .

Gârleanu and Pedersen

When an investor finds an asset manager and confirms that the manager has the technology to obtain the signal, they negotiate the asset management fee f . The fee is set through [Nash bargaining](#) and, at this bargaining stage, both the manager's information acquisition cost and the investor's search cost are sunk.

Gârleanu and Pedersen

Finally, the economy features a group of “noise traders” and a group of “noise allocators”.

Noise traders create uncertainty about the supply of shares and are used in the literature to capture the fact that it can be difficult to infer fundamentals from prices.

We also have the concept of “noise allocators” of total mass $N \geq 0$, who allocate their funds across randomly chosen asset managers. Noise allocators play a similar role in the market for asset management to the one that noise traders play in the market for assets. Noise allocators can make it difficult for searching investors to determine whether a manager is informed by looking at whether they have other investors. Further, the existence of noise allocators changes the performance characteristics across managers and investors.

Gârleanu and Pedersen

There exists a risk-free asset normalised to deliver a zero net return, and a risky asset with payoff v distributed normally with mean $\bar{v}$ and standard deviation σ_v . Agents can obtain a signal s of the payoff, where

$$s = v + \varepsilon$$

The noise ε has mean zero and standard deviation σ_ε , is independent of v , and is normally distributed.

Gârleanu and Pedersen

The risky asset is available in stochastic supply given by q , which is jointly normally distributed with, and independent of, the other exogenous random variables. The mean supply is $\bar{q}$ and the standard deviation of the supply is σ_q . We think of the noisy supply as the number of shares outstanding $\bar{q}$ plus the supply $q - \bar{q}$ from the noise traders.

Gârleanu and Pedersen

Given this asset market, uninformed investors buy a number of shares x_u as a function of the observed price p to maximise their utility u_u (certainty-equivalent wealth), taking into account the fact that the price p may reflect information about the value

$$u_u(W) = -\frac{1}{\gamma} \ln \left(\mathbb{E} \left[\max_{x_u} \mathbb{E} \left[e^{-\gamma(W+x_u(v-p))} | p \right] \right] \right) = W + u_u(0) \equiv W + u_u$$

We can see here how CARA makes the investors decision independent of wealth. Wealth level simply shifts the utility function without affecting their behaviour.

Gârleanu and Pedersen

Asset managers observe the signal and invest in the best interest of their investors. This informed investing gives rise to the gross utility u_i of a active investor. Note that the expectation is now dependent on both the price and the signal

$$u_i(W) = -\frac{1}{\gamma} \ln \left(\mathbb{E} \left[\max_{x_i} \mathbb{E} \left[e^{-\gamma(W+x_i(v-p))} | p, s \right] \right] \right) = W + u_i(0) \equiv W + u_i$$

Note that all investors with an informed manager want the same portfolio x_i since investors are homogeneous. Hence, we simply assume that the manager offers the portfolio x_i for anyone investing W .

Gârleanu and Pedersen

First, we consider the (partial) equilibrium in the asset market given the numbers of informed and uninformed investors. We denote the mass of informed investors by I and note that it is the sum of the number A of rational investors who decide to search for a manager and the number of noise allocators who happen to find an informed manager, where the latter is the total number N of noise allocators times the fraction $M/\bar{M}$ of informed managers

$$I = A + N \frac{M}{\bar{M}}$$

Clearly, the remaining investors, $\bar{A} + N - I$, invest as uninformed, either directly or via an uninformed manager.

Gârleanu and Pedersen

An *asset-market equilibrium* is an asset price p such that the asset market clears:

$$q = Ix_i + (\bar{A} + N - I)x_u$$

Where x_i is the demand that maximises the utility of informed investors given p and the signal s , and x_u is the demand of uninformed investors. The market-clearing condition equates the noisy supply q with the total demand from all informed and uninformed investors.

Gârleanu and Pedersen

Now, we define a *general equilibrium for assets and asset management* as a number of informed asset managers M , a number of active investors A , an asset price p , and asset management fees f such that

- ① No manager would like to change their decision of whether to acquire information
- ② No investor would like to switch status from active (with an associated utility of $W + u_i - c - f$) to uninformed (conferring utility $W + u_u$) or vice-versa
- ③ The price is an asset-market equilibrium
- ④ The asset management fees are the outcome of Nash bargaining

Gârleanu and Pedersen

In a linear equilibrium, an informed agent's demand for is a linear function of prices and signals, and the price is a linear function of the signal and the noisy supply

$$p = \theta_0 + \theta_s((s - \bar{v}) - \theta_q(q - \bar{q}))$$

Where the coefficients θ_0 , θ_s and θ_q are derived in equilibrium. The key property of the price is its *efficiency* (or informativeness) defined as $\frac{\text{var}(v|s)}{\text{var}(v|p)}$. For convenience, we concentrate on the quantity

$$\eta \equiv \ln \left(\frac{\sigma_{v|p}}{\sigma_{v|s}} \right) = \frac{1}{2} \ln \left(\frac{\text{var}(v|s)}{\text{var}(v|p)} \right)$$

Which represents the price *inefficiency*. This quantity records the amount of uncertainty about the asset value for someone who only knows the price p , relative to the uncertainty remaining when one knows the signal s .

Gârleanu and Pedersen

The price inefficiency is linked to investors' value of information. Indeed, η gives the relative utility of investing based on the manager's information (u_i) versus investing as uninformed (u_u)

$$\gamma(u_i - u_u) = \eta$$

The inefficiency η can be written as an explicit function of the number of informed investors I :

$$\eta = -\frac{1}{2} \ln \left(1 - \frac{\sigma_q^2 \sigma_\varepsilon^2}{I^2 / \gamma^2 + \sigma_q^2 \sigma_\varepsilon^2} \frac{\sigma_v^2}{\sigma_v^2 + \sigma_\varepsilon^2} \right) \in (0, \infty)$$

We can see the η is decreasing in I , which is natural since, when there are more informed investors, asset prices become less inefficient (lower η), implying that informed and uninformed investors receive more similar utilities (lower $u_i - u_u$).

Gârleanu and Pedersen

Now we derive the equilibrium asset management fee. The asset management fee is set through Nash bargaining between an investor and a manager. The bargaining outcome depends on each agent's utility in the events of agreement and no agreement.

The investor's utility when agreeing on a fee f is $W - c - f + u_i$. If no agreement is reached, the investor's outside option is to invest as uninformed with his remaining wealth, yielding a utility of $W - c + u_u$. Hence the investor's surplus from agreement is given by

$$S_i = (W - c - f + u_i) - (W - c + u_u) = u_i - u_u - f$$

Similarly, the asset manager's gain from agreement is the fee f . This is true because the manager's information cost k is sunk and there is no marginal cost to taking on the investor. Therefore the asset manager's surplus is given by

$$S_{am} = f$$

Gârleanu and Pedersen

The bargaining outcome maximises the product of the utility gains from agreement, i.e.

$$\max_f S_i S_{am}$$

Which becomes

$$\max_f [u_i - u_u - f] f$$

The solution is the equilibrium asset management fee f given by

$$f = \frac{\eta}{2\gamma}$$

This equilibrium is simple and intuitive: The fee would naturally have to be zero if asset markets were perfectly efficient, so that no benefit of information existed ($\eta = 0$) and increases in the size of the market inefficiency.

Active asset management fees can be viewed as evidence that investors believe that security markets are less than fully efficient.

Gârleanu and Pedersen

We next derive investor's and managers' decisions. An investor optimally decides to look for an informed manager as long as

$$u_i - c - f \geq u_u$$

Recalling the equality $\eta = \gamma(u_i - u_u)$, the investor's optimality condition can be written as

$$\eta \geq \gamma(c + f)$$

This relation must hold with equality in an “interior” equilibrium (i.e. an equilibrium in which strictly positive amounts of investors decide to invest as uninformed and through asset managers — as opposed to all investors making the same decision).

Inserting the equilibrium asset management fee, we get

$$c = \frac{\eta}{2\gamma}$$

Gârleanu and Pedersen

Similarly, we can see that an investor would prefer using an asset manager to acquiring the signal singlehandedly provided that

$$k \geq c + f$$

Using the equilibrium asset management fee, the condition that asset management is preferred to buying the signal can be written as

$$k \geq 2c$$

In other words, finding an asset manager should cost at most half as much as being one, which seems to be a condition that is clearly satisfied in the real world. We can also write equivalently

$$A \geq 2M$$

That is, there must be at least two searching investors for every manager, which is also a realistic implication.

Gârleanu and Pedersen

A prospective informed asset manager must pay the cost k to acquire information. On the other hand, by becoming informed, the manager can expect to have more investors. Specifically, each manager receives a noisy number of investors, but, since managers are risk neutral, they optimise the expected fee revenue net of information costs.

Gârleanu and Pedersen

An uninformed manager expects $N/\bar{M}$ investors, that is, the number of noise allocators divided by the total number of managers. An informed manager expects $A/M + N/\bar{M}$ investors since they expect a fraction of the searching investors in addition to the noise allocators. Therefore, they choose to become informed provided that the expected extra fee revenue covers the cost of information:

$$f \frac{A}{M} \geq k$$

This manager condition must hold with equality for an interior equilibrium, and we can easily insert the equilibrium fee to get

$$M = \frac{\eta A}{2\gamma k}$$

Gârleanu and Pedersen

General Equilibrium for Assets and Asset Management

We focus on interior equilibria, we have arrived at the following two indifference conditions:

$$\frac{\eta(I)}{2\gamma} = c(M, A) \quad [\text{investors' indifference condition}]$$

$$\frac{\eta(I)}{2\gamma} = \frac{M}{A}k \quad [\text{asset managers' indifference condition}]$$

Where η is a function of $I = A + N \frac{M}{M}$ which is given explicitly. Hence, solving the general equilibrium comes down to solving these two explicit equations in two unknowns (M, A) .

Gârleanu and Pedersen

Proposition

In a general equilibrium for assets and asset management:

- ① *Informed asset managers outperform uninformed investing before and after fees, $u_i - f > u_u$. Uninformed asset managers underperform after fees*
- ② *Searching investors' outperformance net of fees just compensates their search costs in an interior equilibrium, $u_i - f - c = u_u$. Larger equilibrium search frictions imply higher net outperformance for informed managers*
- ③ *The asset-weighted average manager (or, equivalently, the asset-weighted average investor) outperforms after fees if and only if the number N of noise allocators is small relative to the number A of searching investors,*
$$A \geq N \left(1 - 2\frac{M}{M}\right)$$

Summary

This lecture discusses how information affects asset prices.

- The idea of a perfectly efficient market results in three puzzles that cannot be explained by the theory of efficient asset markets.
- The evidence also implies that efficient markets may not exist in reality.
- We can account for two of these puzzles and the empirical regularities by incorporating search frictions to the information gathering process as in [Gârleanu and Pedersen \(2018\)](#).

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